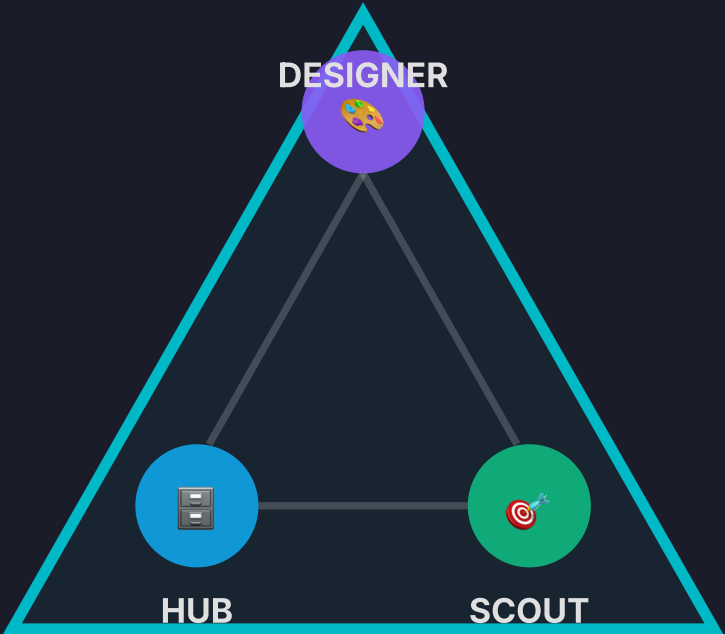




Team Details & Artifact Links



Triangle Smart Targeting
& Real-Time Activation

<epam>
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TriStar

EPAM HACKATHON 2026 · LOYALTY AI PLATFORM

TEAM

Representative: *Ram Naresh, Kartikey Puri*

TEAM MEMBERS



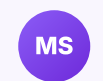
Ram Naresh
AI Architecture & Backend

Rep

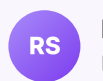


Kartikey Puri
Full Stack & Integration

Rep



Mohit Soni
Frontend & UX Design



Riyaj Shaikh
Backend & DevOps

 **Designer — AI Offer Copilot**

 **Hub — Lifecycle State**

 **Scout — Real-Time Activation**

What we built: AI-powered three-layer loyalty engine — **Designer** turns plain-English objectives into risk-validated OfferBriefs; **Hub** tracks lifecycle in Redis; **Scout** activates at the perfect moment using GPS, weather & behaviour signals. Stack: **React 19 · FastAPI · Claude API · Redis · Azure.**



git.garage.epam.com/kartikey_puri/tristar

EPAM Garage · develop branch · Full source code





Problem Statement

Loyalty programs still behave like batch spreadsheets, not real-time brains — most offers are broad, not truly 1:1, and miss the moment that matters:



~70%

offers go unnoticed*



Hours

lost per campaign



Near 0

real-time context used

- High-value lapsed members still systematically under-targeted
- Most offers ignore real-time context — location, weather, behaviour
- No closed-loop feedback → same campaign mistakes repeated weekly
- Manual, error-prone workflows — mis-configs & margin losses slip through
- Siloed tools (LBM · AEM · SFMC · GL) — no lifecycle view draft-to-expiry



Batch Offers



Member Ignores



Revenue Lost

* Industry benchmark — open + click + redemption drop-off across typical loyalty campaigns

CATEGORY

AI-Powered Loyalty Personalisation

Aligned with Customer Experience & Digital Innovation



Solution

TriStar transforms loyalty into an **AI-powered, real-time engagement engine** — delivering the **right offer** to the **right member** at the **right moment**.

SYSTEM OVERVIEW — END-TO-END FLOW

LAYER 1 · DESIGNER

Biz Objective

Inventory Hint

Live Deal Feed

Claude Sonnet 4.6

OfferBrief JSON

Risk Check

✓ Pass → Hub (draft)

✗ Critical → Block

↓ approved

THE HUB — SHARED CONTEXT STATE

Redis / In-Memory · draft → approved → active → expired · Audit Log · Expiry Sweep 300s

↓ context match

LAYER 3 · SCOUT — REAL-TIME ACTIVATION ENGINE

Flow A

Purchase Event

HMAC webhook · Deterministic ≥ 70

Flow B

Hub Match

Claude AI > 60 · CASL dedup

Flow C

Partner Trigger

Claude Haiku · 24h brief

Flow D

Self-Activation

Customer-accept JWT



Push Notification

Queue for Later (8am)

Activation Log

Tech Stack

Next.js 15

FastAPI

Claude Sonnet 4.6

Redis

Azure

Pydantic v2

Impact

+25%

lapsed member reactivation

80%

faster campaign creation

<200ms

real-time activation

Benefits

80%
Faster
Campaigns

+25%
Member
Reactivation

100%
Audit
Ready

For Marketers

- Generate validated OfferBriefs in seconds — not hours
- Built-in risk detection catches over-discounting before launch
- Inventory-driven recommendations increase campaign ROI
- Real-time risk flags surfaced before any offer goes live

For Members

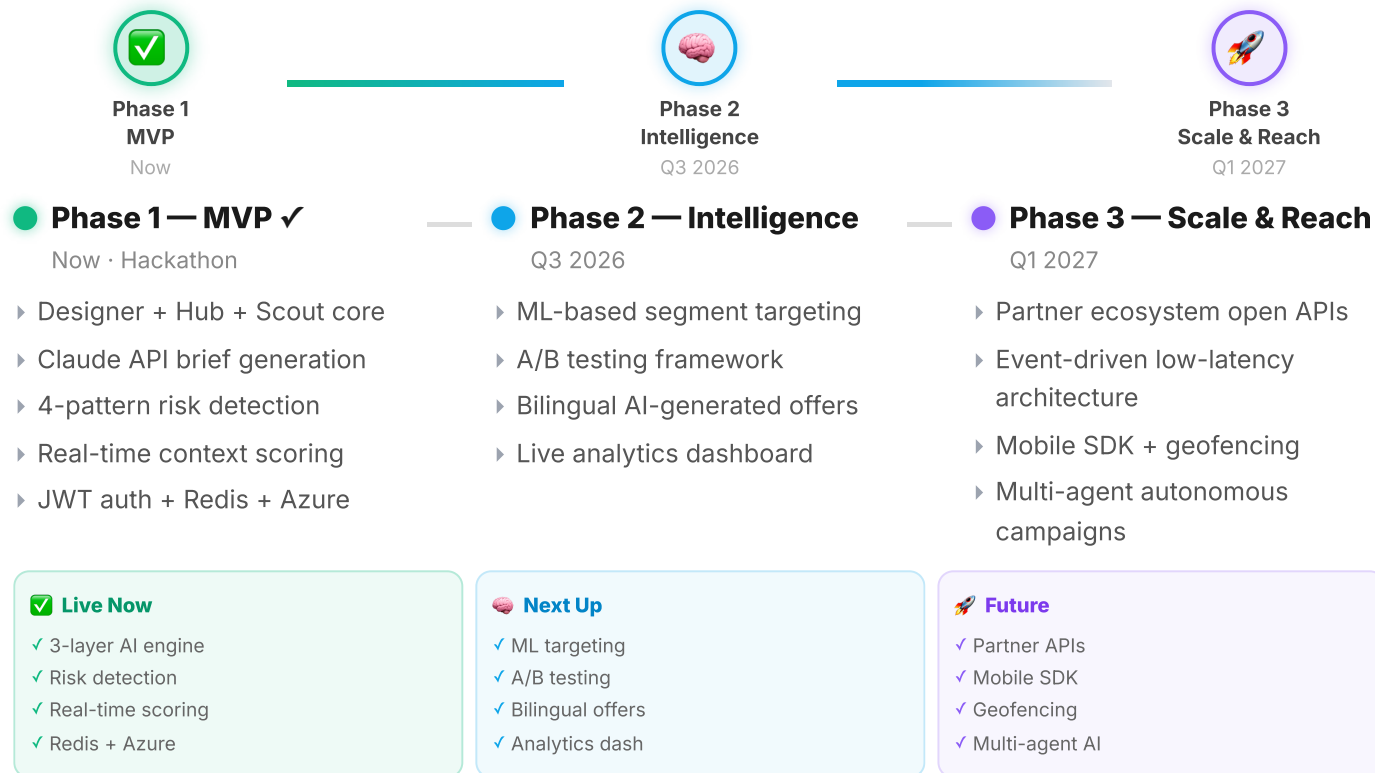
- Offers arrive only when GPS, time & weather align
- Fewer, more relevant notifications — less fatigue
- Seamless, personalised loyalty experience at scale

For the Business

- Full lifecycle visibility: draft → approved → active → expired
- Immutable audit trail — OWASP & compliance ready
- Horizontal scale on Azure · millions of members

Future Enhancements

PRODUCT ROADMAP



Built-in Guardrails (Phase 1)

🔒 1 offer / member / hr

🔒 No duplicates / 24h

🔒 Quiet hrs 10 pm–8 am

🔒 Critical risk → hard block

Security: OWASP Top 10 mitigations built in — parameterised queries, Zod + Pydantic input validation, JWT 1h expiry, Azure Key Vault for secrets, PII-free logs (member_id only).

Demo

Thank You